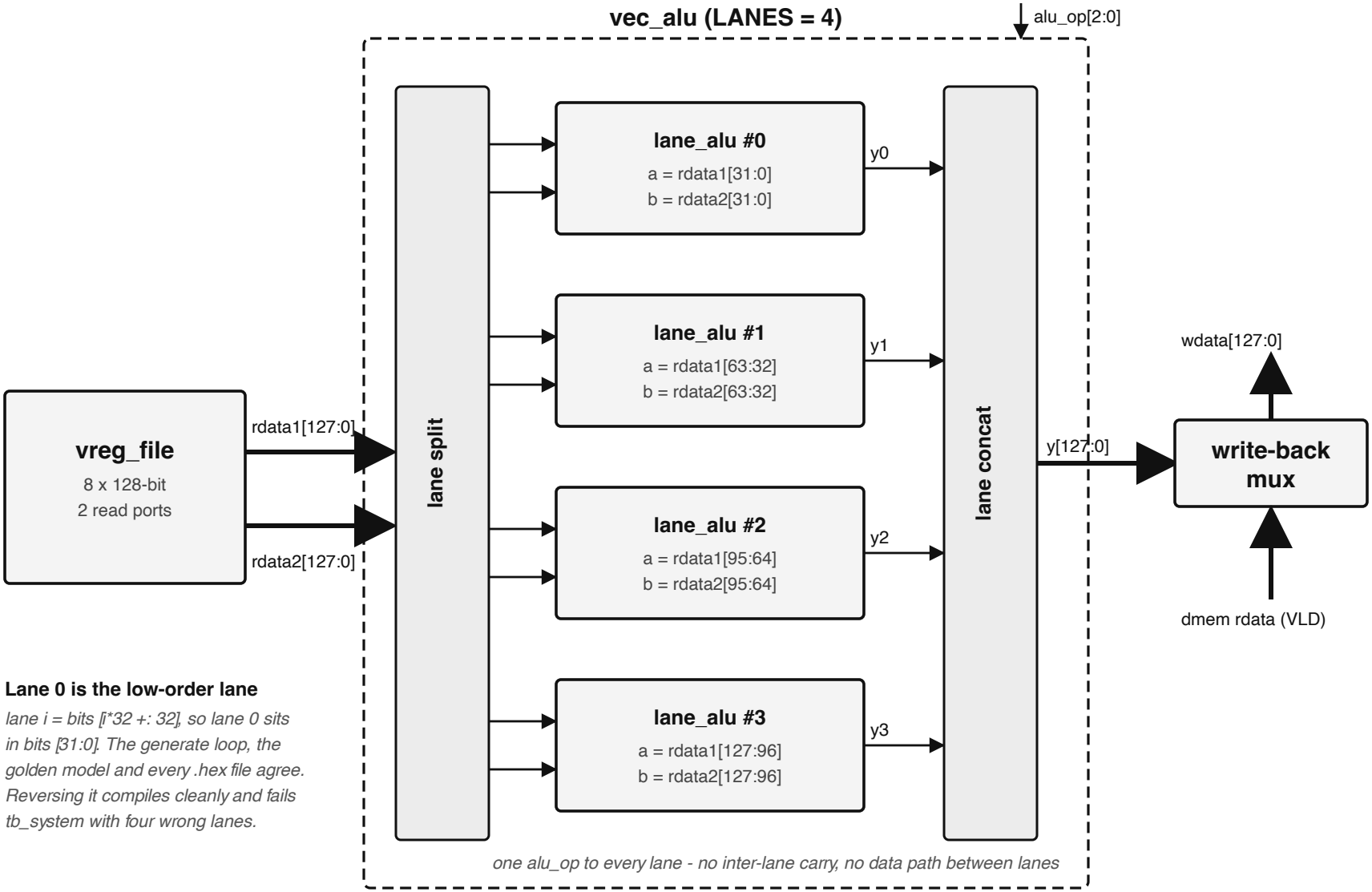

Lane 0 is the low-order lane

*lane i = bits [i*32 +: 32], so lane 0 sits in bits [31:0]. The generate loop, the golden model and every .hex file agree. Reversing it compiles cleanly and fails tb_system with four wrong lanes.*